

Summary of *Robust Agents Learn Causal World Models*

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Basic setup

Let $\mathbf{V} = \{V_1, \dots, V_n\}$ be a finite collection of random variables. A Bayesian network on $\mathbf{V}$ is a pair $M = (G, P)$, where G is a DAG on the V_i and the joint law factorizes according to G :

$$P(v_1, \dots, v_n) = \prod_i P(v_i \mid pa_i),$$

where pa_i denotes the values of the parents $\text{Pa}(V_i)$ of V_i .

In the paper the V_i come in three types: environmental (chance) variables $\mathbf{C} = \{C_1, \dots, C_m\}$, a decision node D , and a utility node U . This models a decision problem: the agent observes the parents $\text{Pa}(D)$ of the decision node and chooses the value of D according to some policy; each outcome carries a utility given by the value of U .

In simplified terms, the paper asks: **under what circumstances does the agent’s family of policies contain enough information to reconstruct the underlying causal model (G, P) ?**

Interventions

The paper’s answer is essentially that the agent must be sufficiently *robust* to certain perturbations of its environment.

The basic perturbation is a *local intervention*. For instance, we may replace the local conditional law at node C_i with a point mass:

$$P_{do(C_i=c'_i)}(c_1, \dots, c_n) = \mathbf{1}_{\{c_i=c'_i\}} \prod_{j \neq i} P(c_j \mid pa_j).$$

Note that $do(\cdot)$ is *not* conditioning under P ; it specifies a new interventional law. In general a local intervention has the form $\sigma = do(C_i = f(C_i))$, under which

$$P(c_i \mid pa_i; \sigma) = \sum_{c'_i: f(c'_i)=c_i} P(c'_i \mid pa_i),$$

with a constant f giving a *hard* intervention. We also allow mixtures $\sigma = \sum_j q_j \sigma_j$ with $\sum_j q_j = 1$, meaning σ_j is applied with probability q_j , so that $P(\cdot \mid \sigma) = \sum_j q_j P(\cdot \mid \sigma_j)$.

A policy is a conditional law $\pi(d \mid pa_D)$ over decisions given the observed variables, and the utility is a real-valued function $U = U(pa_U)$. An optimal policy satisfies $\pi^* \in \arg \max_{\pi} \mathbb{E}^{\pi}[U]$.

The paper assumes an *unmediated decision task*, $\text{Desc}(D) \cap \text{Anc}(U) = \emptyset$: the decision influences utility only directly (typically $D \in \text{Pa}(U)$), never through intermediate environmental variables.

Domain shifts and regret

The paper studies generalization beyond the i.i.d. setting. Rather than being tested under the training law P , the environment may shift to an intervention-induced law P_{σ} , while the utility function stays fixed. These are the *domain shifts*.

Under a shift σ , let π_σ^* be the optimal policy and π_σ the agent's policy. Its regret is

$$\text{Regret}_\sigma(\pi_\sigma) = \mathbb{E}_{\pi_\sigma^*}[U] - \mathbb{E}_{\pi_\sigma}[U],$$

and the policy is δ -optimal if $\text{Regret}_\sigma(\pi_\sigma) \leq \delta$. Regret is an external measure: the agent need not know the optimum.

A second assumption is *domain dependence*: there must exist compatible environment distributions under which the optimal policy differs. Otherwise the policy is the same everywhere and cannot reveal anything about the environment. The central object is therefore the family $\{\pi_\sigma\}_{\sigma \in \Sigma}$, where Σ is the class of mixtures of local interventions.

Main theorems

Theorem 1 (exact reconstruction). For almost all causal influence diagrams satisfying the assumptions above, knowing the optimal policy $\pi_\sigma^*(d \mid pa_D)$ for every $\sigma \in \Sigma$ determines the causal graph and the local conditional laws on the ancestors of the utility:

$$\{\pi_\sigma^*\}_{\sigma \in \Sigma} \implies (G, P) \text{ on } \text{Anc}(U).$$

The exceptional cases are degenerate parameter choices for which some causal relation never affects expected utility, and is therefore invisible to optimal decisions.

Theorem 2 (approximate reconstruction). If $\text{Regret}_\sigma(\pi_\sigma) \leq \delta$ for all $\sigma \in \Sigma$, then one can reconstruct an approximate model $M' = (G', P')$ with

$$|P'(v_i \mid pa_i) - P(v_i \mid pa_i)| \leq \gamma(\delta), \quad \gamma(0) = 0, \quad \gamma(\delta) = O(\delta)$$

for small δ . Weak causal edges may be missed when their effect on expected utility falls below the regret scale.

Theorem 3 (converse). If (G, P) is known, one can compute an optimal policy under every allowed intervention; and if an approximate model has local probability error at most ε , the induced policy has regret $\delta = O(\varepsilon)$. Schematically,

$$\text{accurate causal model} \iff \text{low-regret adaptation across domain shifts.}$$

Reconstruction from policy behaviour

The key idea is to turn the points at which the policy *switches* into numerical causal information. Suppose interventions σ_1, σ_2 favour different decisions d_1, d_2 , and form the mixture $\sigma(q) = q\sigma_1 + (1 - q)\sigma_2$. Expected utility is affine in q :

$$\mathbb{E}[U \mid d, \sigma(q)] = q \mathbb{E}[U \mid d, \sigma_1] + (1 - q) \mathbb{E}[U \mid d, \sigma_2].$$

Set $\Delta_i = \mathbb{E}[U \mid d_1, \sigma_i] - \mathbb{E}[U \mid d_2, \sigma_i]$. At the critical value q_* where the optimal action switches, $q_*\Delta_1 + (1 - q_*)\Delta_2 = 0$, so

$$\Delta_1 = -\frac{1 - q_*}{q_*} \Delta_2.$$

If σ_2 is a hard intervention fixing the relevant variables, then Δ_2 is known directly from the utility function. Hence the observed switching point q_* reveals the unknown utility difference Δ_1 .

Such utility differences are linear functions of the unknown interventional probabilities. Repeating the construction yields probabilities such as $P(C_i = c_i \mid do(\mathbf{C} \setminus C_i))$, from which one reads off the parent sets and the local conditional laws $P(C_i \mid \text{Pa}(C_i))$. Thus

$$\text{policy switch} \rightarrow \text{utility difference} \rightarrow \text{interventional probability} \rightarrow (G, P).$$

Interpretation

The theorem does not say that the agent literally stores a symbolic DAG. Rather, if it stays near-optimal across a sufficiently rich family of intervention-induced domains, then its policy behaviour contains enough information for an *external observer* to reconstruct the utility-relevant causal model. In this information-theoretic sense,

$$\boxed{\text{robust OOD behaviour} \implies \text{implicit causal knowledge.}}$$

Two limits are worth keeping in view for discussion. First, this is an identifiability result, not a statement about learning: it says the causal model is *recoverable from* a robust policy, not that any particular training procedure produces one. Second, it is scoped to unmediated tasks and to shifts generated by (mixtures of) local interventions; how far the conclusion survives mediated decisions or richer shift classes is left open.